

Cracking of High Density Polyethylene Pyrolysis Waxes on HZSM-5 Catalysts of Different Acidity

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ABSTRACT: High density polyethylene (HDPE) cracking has been carried out in a thermal-catalytic two-step unit for the selective production of light olefins. Continuous pyrolysis of HDPE has been conducted in a conical spouted bed reactor at 500 °C, and the volatiles formed (mainly waxes) have been transformed in a downstream fixed bed catalytic reactor at 500 °C. The effect of catalyst acidity on product yield and composition has been studied by using three catalysts based on HZSM-5 zeolites with a SiO₂/Al₂O₃ ratio of 30, 80, and 280. The maximum light olefin yield (58 wt %) has been obtained using the most acidic catalyst (SiO₂/Al₂O₃ ratio of 30), with the individual yields of ethylene, propylene, and butenes being 9.5, 32, and 16.5 wt %, respectively. The results are a clear evidence of the higher efficiency of the two-step reaction system compared to the in situ catalytic pyrolysis (single-step), which is explained by the suitable combination of operating conditions in each one of the steps.

1. INTRODUCTION

The increase in the production and consumption of plastic materials and the environmental issues related to their low biodegradability lead to the need for a decrease in plastic waste landfill and the promotion of recycling.^{1,2} Feedstock recycling methods have been considered not only the most feasible at a large scale but also economically viable and environmentally friendly.^{3,4}

The pyrolysis process is considered as a stand-alone facility for the valorization of plastics, especially polyolefins (²/₃ of waste plastics), in order to obtain liquid or gaseous fuels or raw chemicals, such as light olefins and BTX. Accordingly, polyolefin pyrolysis has attained a significant development stage based on several reactor types, that is, rotating furnace, tubular, rotating cone and screw reactors, and fluidized bed and spouted bed reactors.⁵ Pyrolysis can be carried out in small units and close to the collection points, thus avoiding the costs involved in the transportation of plastic wastes. The use of acid catalysts decreases the activation energy and temperature required for the pyrolysis process, and their selection is based on the operating conditions and products of interest, such as olefins or gasoline.^{6,7}

HZSM-5 zeolites have been studied by several researchers to produce a product stream composed of mainly light olefins or low aromatic content gasoline.^{8–13} Elordi et al. enhance the access of macromolecules into the HZSM-5 zeolite microporous structure by generating mesopores in the catalyst particle by agglomeration of the zeolite with bentonite and alumina.^{14–16} Furthermore, the shape selectivity in the microporous structure of the HZSM-5 limits the secondary reactions of hydrogen transfer and polyaromatic generation, thus contributing to an increase in the light olefin selectivity.¹⁷ In addition, the three-dimensional structure facilitates coke precursor diffusion toward the outside of the zeolite structure, which is enhanced by the high N₂ flow rate in the pyrolysis process, thus leading to a limited deactivation.^{18,19}

A suitable reactor is essential for plastic waste pyrolysis because of the sticky nature and low thermal conductivity of the wastes.²⁰ The plastic must melt and coat the bed particles prior

to devolatilization. This process requires high heat and mass transfer rates in order to operate under isothermal conditions with short volatile residence times and, therefore, hindered byproduct formation. Although fluidized beds have been commonly used for plastics pyrolysis,^{21–28} the conical spouted bed reactor (CSBR) performs satisfactorily in the pyrolysis of polyolefins,^{14–16,29} given that the vigorous cyclic movement of the sand particles avoids the defluidization problems commonly encountered in fluidized beds.²⁶ Furthermore, the cyclic movement of the bed particles and the high inert gas (nitrogen) flow rate enhance heat and mass transfer between phases and reduce the residence time of the volatile stream (to the order of centiseconds).

The features of the conical spouted bed reactor (CSBR) make it especially suitable for the selective production of waxes (C₂₁₊) from high-density polyethylene (HDPE) pyrolysis at relatively low temperatures (500 °C), with yields being high (67 wt %) and operation occurring without defluidization problems.²⁹ The cyclic movement of the particles in the bed promotes a homogeneous coating of sand/catalyst particles with the fused HDPE and the efficient gas–solid contact favors heat and mass transfer between phases.³⁰ Consequently, the volatile stream leaving the CSBR reactor (mainly waxes) is suitable for downstream upgrading for the production of valuable products.

A two-step unit allows the use of different operating conditions in each step, thus maximizing light olefin production. Accordingly, San Miguel et al.³¹ studied the effect of temperature (425–475 °C) on product distribution in a catalytic fixed-bed reactor (with HZSM-5, H β , or Al-MCM-41 zeolites) located downstream from the pyrolytic batch reactor (operating at 425 °C). Recently, Artetxe et al.^{32,33} performed the downstream thermal and catalytic cracking in continuous

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Table 1. Physical Properties of HZSM-5-Based Catalysts

SiO ₂ /Al ₂ O ₃ ratio	BET surface area (m ² g ⁻¹)	micropore ($d_p < 20$ Å) area (m ² g ⁻¹)	mesopore ($20 < d_p$ (Å) < 500) volume (cm ³ g ⁻¹)	pore volume distribution (%) (<20/20 < d_p (Å) < 500/>500)	average pore diameter (Å)
30	182	98	0.26	4.5/27.5/68.0	84
80	166	100	0.11	6.3/16.4/77.3	39
280	170	92	0.16	5.7/19.1/75.2	47

mode of the waxes obtained in a CSBR, and attained high olefin yields. The thermal treatment was carried out in a multitubular reactor operating at 900 °C, with C₂–C₄ olefin production being 76 wt % at this temperature.³² The catalytic cracking was carried out in a fixed-bed reactor with a HZSM-5 catalyst, and the maximum yield of light olefins was 62.9 wt % by operating at 550 °C.³³

This paper develops the selective production of light olefins from HDPE by means of a two-step thermal and catalytic process. The volatiles formed in the pyrolysis of the plastic in a CSBR (mainly waxes) are catalytically converted in a fixed-bed reactor on a HZSM-5 zeolite-based catalyst. The influence of the acid properties of the zeolite (SiO₂/Al₂O₃ ratio) on product yield and composition has been studied in order to maximize light olefins yield.

2. EXPERIMENTAL SECTION

2.1. Raw Material and Catalysts. The high-density polyethylene (HDPE) was supplied by Dow Chemical (Tarragona, Spain) in the form of cylindrical pellets (4 mm), with the following properties: average molecular weight, 46.2 kg mol⁻¹; polydispersity, 2.89; and density, 940 kg m⁻³. The higher heating value, 43 MJ kg⁻¹, has been measured by differential scanning calorimetry (Setaram TG-DSC-111) and isoperibolic bomb calorimetry (Parr 1356).

Three catalysts have been prepared based on HZSM-5 zeolite with SiO₂/Al₂O₃ ratios of 30, 80, and 280, provided by Zeolyst International (Kansas City, USA). The zeolites have been calcined at 550 °C in order to obtain the acid form, as they have been supplied in ammonium form. The zeolites (25 wt %) have been agglomerated by wet extrusion with bentonite (Exaloid, 30 wt %) and inert alumina (Martinswerk, 45 wt %) in order to obtain particles with a suitable size, mechanical resistance, and thermal conductivity.³⁴ Furthermore, the agglomeration generates mesopores and macropores in the catalyst particles, which attenuate the deactivation as coke deposition is promoted on the outside of the pores, thus minimizing external blockage.¹⁸ Prior to use, the catalyst has been calcined at 575 °C for 2 h to eliminate the strong acid sites (hydrothermally unstable) of the HZSM-5 zeolite by dehydroxylation. This catalyst equilibration moderates acid strength, thereby minimizing the secondary reactions of hydrogen transfer involving olefins to yield paraffins, aromatics and coke, and therefore reducing catalyst deactivation. Furthermore, the catalyst fully recovers its kinetic behavior when it is regenerated by coke combustion with air at 550 °C, which allows using it in reaction–regeneration cycles.³⁵

The catalysts have been characterized by measuring their physical and acid properties. The physical properties (BET surface area, average pore diameter, and pore volume distribution), Table 1, have been measured by N₂ adsorption–desorption (Micromeritics ASAP 2010). Figure 1 shows the N₂ adsorption–desorption isotherms of the catalysts used. As observed, the physical properties of the three catalysts are similar, as they have the same meso- and macroporous structure

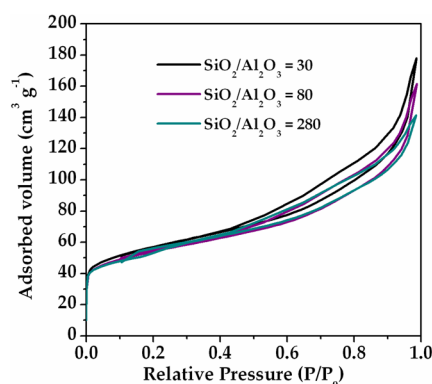


Figure 1. Adsorption isotherms of the catalysts.

provided by the agglomeration with bentonite and alumina. Consequently, their higher mesopore volume and average pore diameter than the parent zeolite is due to the effect of SiO₂/Al₂O₃ ratio in the formation of zeolite particle agglomerates. Thus, a low SiO₂/Al₂O₃ ratio (30) favors the formation of mesopores between zeolite particles in the catalyst preparation stage by agglomeration with bentonite and alumina.

The acid properties of the catalysts have been determined by NH₃ adsorption–desorption: the values of total acidity and average acid strength have been obtained by monitoring the differential adsorption of NH₃ simultaneously by calorimetry and thermogravimetry in a Setaram TG-DSC 111 and the curve for temperature programmed desorption of NH₃ has been obtained by connecting a Blazer Instruments mass spectrometer (Thermostar) online to a Setaram TG-DSC 111.³⁶

Figure 2 displays the acid strength distribution of the three catalysts. The total acidity of the catalyst decreases when the

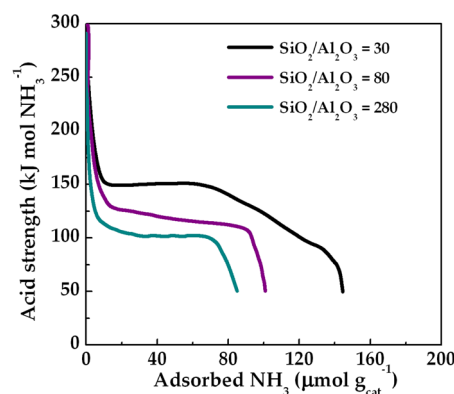


Figure 2. Acid strength distribution of the catalysts.

SiO₂/Al₂O₃ ratio of the zeolite is higher, from 145 μmol_{NH₃} g_{cat}⁻¹ for a ratio of 30 to 85 μmol_{NH₃} g_{cat}⁻¹ for 280. Similarly, the acid strength of the catalyst decreases for a higher SiO₂/Al₂O₃ ratio, from 150 kJ mol_{NH₃}⁻¹ for a ratio of 30 to 100 kJ mol_{NH₃}⁻¹ for 280. The acid sites need to have enough strength

for hydrocarbon conversion, which can be assessed by taking into account that SiO_2 (not active for these reactions) has a NH_3 adsorption value of 40 kJ mol^{-1} at 150°C and strong acid sites require above 100 kJ mol^{-1} for alkane cracking.³⁷

The curves of temperature programmed desorption of NH_3 for the three catalysts are shown in Figure 3. Total acidity (area

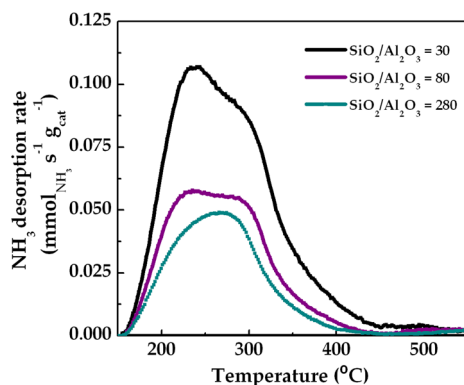


Figure 3. Ammonia TPD curves of the catalysts.

under the curve) increases by decreasing the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of the zeolite. Furthermore, two types of acid sites can be observed: weak sites on which NH_3 is desorbed at 240°C , and sites with moderate acidity on which NH_3 is desorbed at 300°C . The analysis of the acid properties allows the conclusion that all the catalysts are mainly composed of moderate and weak acid sites and that the total acidity and, to a lower extent, the average acid strength of the catalysts decrease when the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of the HZSM-5 zeolite is increased. This moderate acid strength is a consequence of the catalyst calcination at high temperature (575°C) in order to minimize the secondary reactions and enhance its application in reaction–regeneration cycles as has been previously explained.

2.2. Experimental Equipment and Conditions. A scheme of the bench-scale unit used for pyrolysis and catalytic cracking is shown in Figure 4.

The HDPE is fed into the reactor by means of a vessel equipped with a vertical shaft connected to a piston placed below the material bed, which is raised while the whole system is vibrated by an electric motor. The plastic feed rate can be varied from 0.2 g min^{-1} to 5 g min^{-1} . The pipe that connects the feeding system with the reactor is water-cooled, in order to prevent plastic melting and blocking the system. Additionally, a very small N_2 flow rate introduced into the vessel stops the volatile stream entering the feeding vessel.

The pyrolysis reactor is a conical spouted bed (CSBR), which has been specifically designed for handling sand particles coated with melted plastic. The reactor guarantees bed stability in a wide range of operating conditions,³⁸ and the vigorous solid circulation in the reactor ensures bed isothermicity and high heat transfer rates.³⁰ The dimensions of the CSBR reactor (which have been described elsewhere³²) are based on the knowledge acquired in previous hydrodynamic studies,^{39,40} and in the pyrolysis of different plastic materials, such as polystyrene,⁴¹ polymethylmethacrylate,⁴² and polyethylene terephthalate,⁴³ scrap tires,⁴⁴ and lignocellulosic biomass.⁴⁵ The bed was fluidized by means of nitrogen, whose flow rate is controlled by a mass flow controller that allows feeding up to 5 L min^{-1} .

The pyrolysis volatiles (mainly waxes) formed in the CSBR reactor flow toward the second catalytic fixed-bed reactor through a thermostatted line. The fixed bed is a cylindrical stainless steel reactor, with an internal diameter of 13.1 mm and a total length of 305 mm. The second reactor, together with a high efficiency cyclone that retains the fine sand particles entrained from the bed, is placed in a forced convection oven kept at 270°C to avoid the condensation of pyrolysis products. The products formed in the catalytic step circulate through a volatile condensation system consisting of a cool water condenser, a Peltier cooler, and a coalescence filter to ensure the total condensation of volatile hydrocarbons.

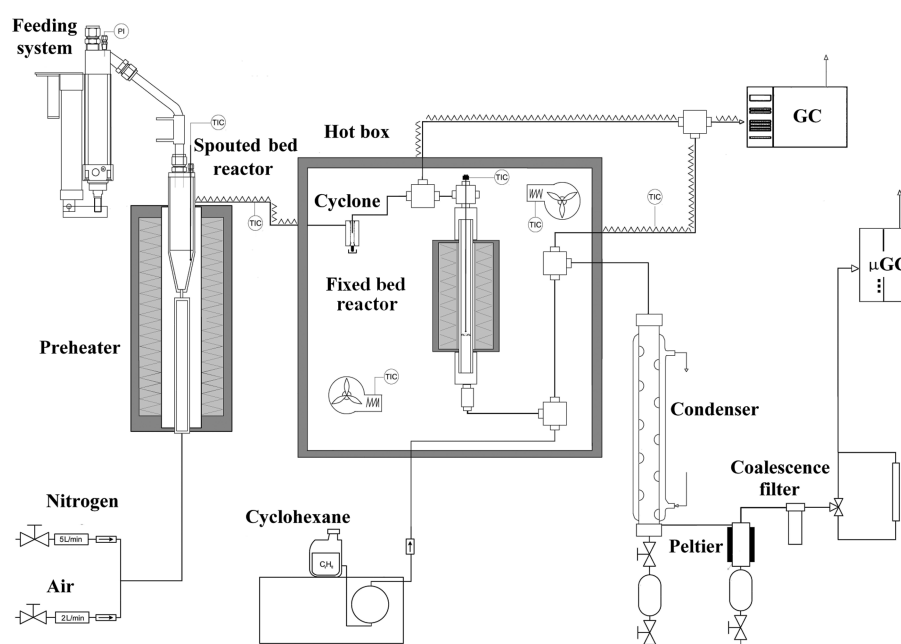


Figure 4. Scheme of the bench-scale plant used for the two-step thermal and catalytic process.

The runs were carried out in continuous regime by feeding 1 g min⁻¹ of HDPE. The CSBR contains 50 g of sand (particle diameter in the 0.3–0.4 mm range). The nitrogen flow rate (measured at room conditions) is 5 L min⁻¹, which is 20% above that corresponding to the minimum spouting velocity.³⁸ The CSBR reactor temperature is kept at 500 °C in all the runs. The average residence time of the volatile product stream ranges from approximately 30 ms in the spout zone to 500 ms in the annulus. The fixed bed consists of 8 g of catalyst (1–2 mm), corresponding to a space–time of 8 g_{cat} min g_{HDPE}⁻¹. The reactor is maintained at 500 °C, and the residence time of the volatiles is around 30 ms. As reported in a previous paper, these operating conditions have been proven to be suitable for light olefin production.³³ A continuous run was carried out for 5 h to obtain enough liquid for its subsequent analysis and qualitatively assess the significance of catalyst deactivation by coke deposition.

2.3. Product Analysis. The product stream leaving the pyrolysis and catalytic reactors were analyzed by an online Varian 3900 chromatograph provided with a HP-Pona column and flame ionization detector (FID), which was connected through a line thermostatted at 280 °C to avoid the condensation of heavy compounds and to quantify all the volatile products. Furthermore, the noncondensable gases were analyzed by means of a micro GC Varian 4900, which allowed a detailed quantification of the product stream.

Cyclohexane (not formed in the process) was used as an internal standard to validate the mass balance, which was fed into the product stream at the outlet of the catalytic reactor (0.05 mL min⁻¹). To ensure the reproducibility of the results, the online GC analysis of the product stream was repeated twice for each time on stream, with the differences observed being below 5%. The mass balance closure for carbon and hydrogen was carried out by monitoring the polymer fed into the reactor and the hydrocarbons at the outlet of the two-step process. Overall mass balance closure is above 95 wt % in all the runs.

The identification of the liquid products was performed in a gas chromatography–mass spectrometry (GC–MS) device (Shimadzu UP-2010S), provided with a HP-Pona column. The gaseous compounds were identified by means of a micro-GC connected to a MS (Agilent 5975B) provided with four modules, with only OV-1 (for nonpolar compounds) and Stabilwax (special for polar compounds) being connected to the mass spectrometer.

3. RESULTS AND DISCUSSION

3.1. Pyrolysis of HDPE at 500 °C (First Step). The pyrolysis of HDPE at 500 °C in a conical spouted bed reactor leads to a product stream composed mainly of waxes (C₂₁₊) given that the low residence time of the gases and the high heat and mass transfer rates between phases in this reactor promote the formation of long chain hydrocarbons by random scission mechanisms. This temperature is suitable for minimizing the energy requirements, as lower temperatures cause particle agglomeration and bed defluidization due to the low pyrolysis reaction rate, whereas higher temperatures result in a sharp increase in the yield of light fractions, with the gaseous products (C₄) prevailing at temperatures above 675 °C.²⁹

Table 2 shows the composition of the volatile stream, in which six groups have been considered: light olefins (C₂–C₄), light alkanes (C₁–C₄), nonaromatic C₅–C₁₁, aromatic C₆–C₁₁, C₁₂–C₂₀ hydrocarbons (nonaromatics), and waxes (C₂₁₊). The

Table 2. Yields (wt %) of Product Fractions Obtained in the Pyrolysis of HDPE in a CSBR at 500 °C

fractions	yield (wt %)
Light olefins (C ₂ –C ₄)	1.15
Ethylene	0.08
Propylene	0.50
Butenes	0.57
Light alkanes (C ₂ –C ₄)	0.35
Methane	0.03
Ethane	0.07
Propane	0.08
Butanes	0.18
Nonaromatic C ₅ –C ₁₁	5.58
Paraffins	0.34
Isoparaffins	2.50
Naphthenes	0.19
Olefins	2.56
Aromatic C ₆ –C ₁₁	0.28
C ₁₂ –C ₂₀	25.64
Diolefins	3.22
Olefins	13.07
Paraffins	9.35
Waxes (C ₂₁ +)	67.0
C ₂₁ –C ₄₀	29.5
C ₄₀ +	37.5

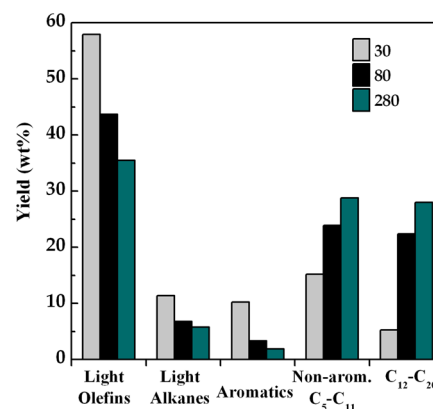


Figure 5. Effect of the SiO₂/Al₂O₃ ratio of the HZSM-5 zeolite on product fraction yields.

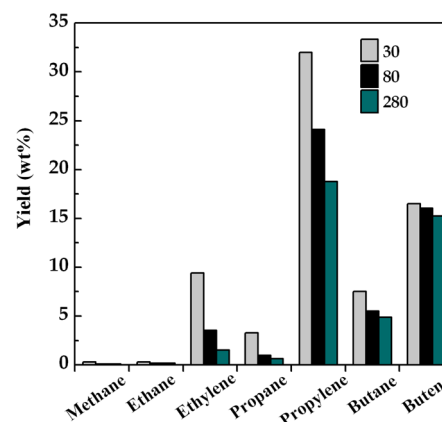


Figure 6. Effect of the SiO₂/Al₂O₃ ratio of the HZSM-5 zeolite on the individual yields of the gaseous compounds.

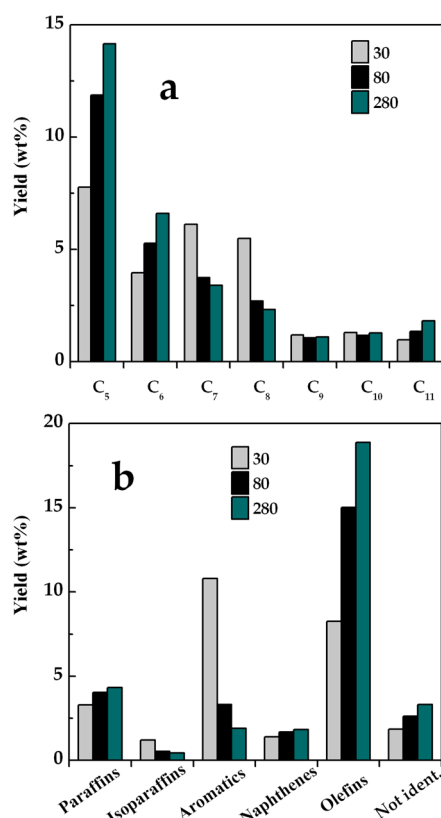


Figure 7. Effect of the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of the HZSM-5 zeolite on the yields of the different fractions in the gasoline group (C_5 – C_{11}) according to the number of carbon atoms (a) and chemical bonds (b).

Table 3. Fuel Properties of the Gasoline Fractions Obtained with the Three Catalysts Based on HZSM-5 Zeolites Having Different $\text{SiO}_2/\text{Al}_2\text{O}_3$ Ratios

$\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio	octane number	olefins (vol %)	aromatics (vol %)	benzene (vol %)
30	94.1	33.1	43.3	4.2
80	86.7	61.2	13.5	1.3
280	85.9	68.9	6.9	0.46
required	95	<18	<35	<1

Table 4. Comparison of the Olefin and Propylene Yields Obtained in the Single-Step Process¹⁶ with Those Obtained in the Two-Step Process

process	operating conditions	$\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio	light olefin yield (wt %)	propylene yield (wt %)
catalytic pyrolysis (single-step)	500 °C, 30 $\text{g}_{\text{cat}} \text{ min}^{-1} \text{ g}_{\text{HDPE}}^{-1}$	30	57.0	28.0
		80	59.8	29.0
pyrolysis + catalytic cracking (two-step)	500 °C + 500 °C, 8 $\text{g}_{\text{cat}} \text{ min}^{-1} \text{ g}_{\text{HDPE}}^{-1}$	30	58.0	32.0
		80	43.6	24.1
		280	35.5	18.8

yields obtained are similar to those obtained by Elordi et al. in a previous paper in which the effect of temperature on product distribution was studied in the 500–700 °C range.²⁹

Waxes are the main compounds with a yield of 67 wt %, followed by C_{12} – C_{20} nonaromatic hydrocarbons (25.6 wt %). A detailed characterization of the waxes can be found

elsewhere.⁴⁶ The yield of the remaining products obtained (nonaromatic C_5 – C_{11} hydrocarbons, light olefins and alkanes) is very low and, furthermore, they may be upgraded by cracking, except methane and aromatics. Consequently, the volatiles formed in this first pyrolysis step are suitable for a further upgrade to light olefins by catalytic cracking. In addition, continuous operation allows the acquisition of a constant stream at the outlet of the pyrolysis step, which is essential to ensure a homogeneous final product stream.

3.2. Effect of Catalyst Acidity on the Catalytic Cracking of Pyrolysis Volatiles (Second Step). The downstream catalytic cracking fully transforms the pyrolysis waxes into lighter compounds, mainly olefins, as shown in Figure 5 for the three catalysts studied. The catalytic cracking of polyolefins takes place via two carbocationic mechanisms: β scission or classic bimolecular mechanism and the proteolytic or monomolecular mechanism through carbonium ions.⁴⁷ The prevailing mechanism depends on the number of carbon atoms in the reactant, the type of bonds (olefinic or paraffinic), the reaction conditions, and the shape selectivity and acid properties of the catalyst. Accordingly, the severe shape selectivity, moderate acid strength, and low hydrogen transfer capacity of the HZSM-5 zeolite are suitable properties for enhancing the monomolecular cracking of polyolefins and minimizing bimolecular reactions, such as olefin oligomerization–cracking to produce paraffins and olefin cyclization and condensation to form coke.^{48,49}

An analysis of the role of the HZSM-5 zeolite acidity, Figure 5, shows that the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of the HZSM-5 zeolite highly affects the product fraction yields. The highest acidic catalyst ($\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of 30) is the most active for cracking waxes, giving way to a higher yield of light olefins and especially a lower yield of the C_{12} – C_{20} heavy fraction, which is also transformed.

The reduction in the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio from 280 to 30 results in an increase in the yield of light olefins from 35.5 to 58 wt % and a decrease in the yields of C_{12} – C_{20} and C_5 – C_{11} fractions from 28 to 5.3 wt % and from 28.8 to 15.2 wt %, respectively. In view of the high yields obtained for the C_{12} – C_{20} fraction on the catalysts with an $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio > 30, it can be concluded that the total acidity of the catalysts is not enough for fully cracking this fraction. Therefore, a higher space–time is needed for the transformation of heavy compounds into olefins on these catalysts.

Furthermore, a decrease in the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of the zeolite increases the yields of light alkanes and aromatics. Thus, catalysts with higher total acidity and average acid strength (Figures 2 and 3) enhance the cracking of C_5 – C_{11} and C_{12} – C_{20} hydrocarbons, but they also promote the secondary reactions of hydrogen transfer and condensation reactions (Diels–Alder reactions) involving light olefins.

The results obtained confirm the suitability of the two-step system proposed for light olefin production from HDPE. The suitable properties of the HZSM-5 zeolite and excellent performance of the reaction system allow minimization of the undesired secondary reactions. Thus, the low residence time of the volatiles in the thermal and catalytic steps and their low concentration in the fixed bed catalytic reactor (diluted by the high nitrogen flow rate used in the pyrolysis step) enhance the selectivity to olefins. Additionally, the low reaction temperatures used also help minimize methane yield, by hindering olefin overcracking reactions.

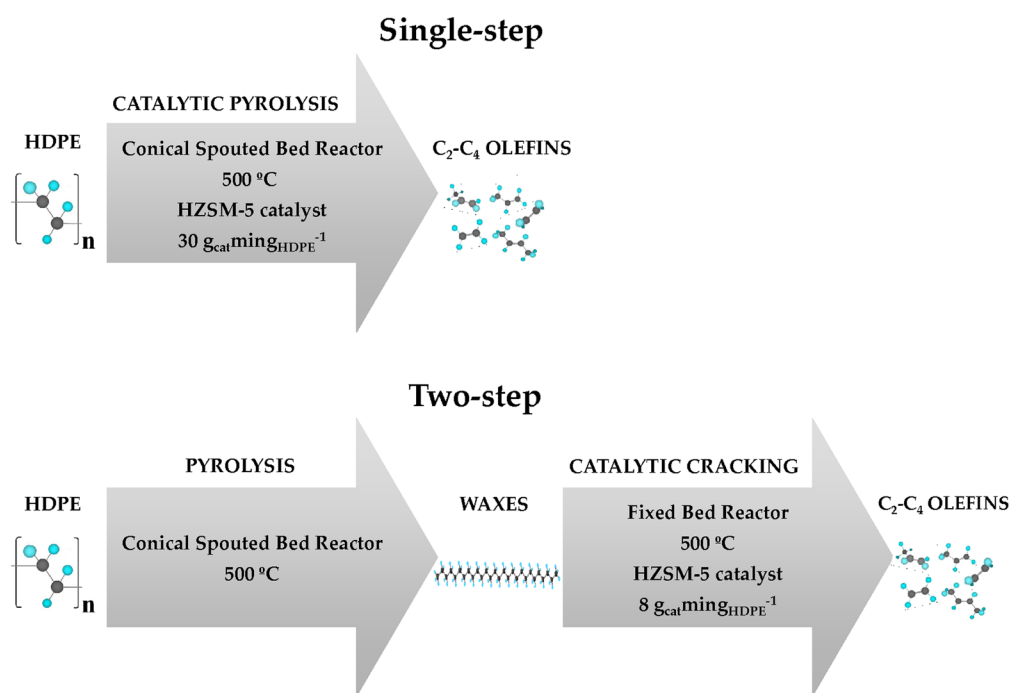


Figure 8. Comparison of the single-step process with the two-step process for light olefin production from HDPE.

Furthermore, catalyst deactivation is low, as the content does not exceed 1 wt % (by mass unit of the whole catalyst, not only the zeolite phase) for five-hour operation (300 g of HDPE), even with the most acidic catalyst. The operating conditions mentioned above allow full conversion of waxes (main coke precursors), and the suitable properties of the HZSM-5 zeolite for improving the flow of coke precursors toward the outside of the zeolite crystals give way to slow deactivation of the catalyst and, therefore, a more viable process. It should be noted that after 5 h of reaction, deactivation is related to coke deposition due to wax condensation mainly on the outside of the zeolite crystals. Coke deposition for longer times on stream will take place presumably inside the zeolite particles, being promoted by condensation reactions involving olefins and aromatics (catalyzed by acid sites). The importance of catalyst deactivation for process feasibility prompts the need for a detailed study on coke deposition, developing a kinetic model for the quantification of the deactivation and the determination of the optimum conditions for minimizing this problem.

Figure 6 displays the individual yields of the products making up the gaseous fraction (light olefins and light alkanes). Propylene is the main compound for the three catalysts studied, and its yield increases as the acidity of the zeolite is increased, from 18.8 to 32 wt % for SiO₂/Al₂O₃ ratios of 280 and 30, respectively. The yield of butenes increases to a minor extent, from 15.3 to 16.5 wt %, and that of ethylene from 1.5 to 9.4 wt %, in the SiO₂/Al₂O₃ range studied. A higher total acidity of the catalyst enhances olefin oligomerization-cracking reactions involving light olefin interconversion, which increases ethylene selectivity.⁵⁰ Therefore, ethylene/propylene ratio increases from 0.08 with a SiO₂/Al₂O₃ ratio of 280 to 0.29 with a ratio of 30.

The influence of the SiO₂/Al₂O₃ ratio on the individual yields of light alkanes is not as significant as on the yields of olefinic compounds. Nevertheless, the higher the acidity of the catalyst is, the higher is the production of light alkanes due to the enhancement of cracking and secondary hydrogen transfer

reactions. Accordingly, the yield of butanes and propane increases to 7.5 and 3.3 wt %, respectively, for a SiO₂/Al₂O₃ ratio of 30. The yields of methane and ethane are very low for the three catalysts studied, which is attributed to the low residence time in the reactor. However, a slightly higher formation is observed on the most acidic catalyst, which indicates that although these compounds are mainly produced by thermal cracking, the catalyst also plays a role in their production.

The gasoline fraction (including both nonaromatic C₅–C₁₁ and single-ring aromatic compounds) was analyzed based on the identification of the main compounds. Figure 7 shows the effect of the SiO₂/Al₂O₃ ratio of the HZSM-5 zeolite on the yields of gasoline compounds ordered according to the number of carbon atoms (Figure 7a) and type of bond (Figure 7b). An increase in catalyst acidity leads to a reduction in the yield of C₅ and C₆ olefinic fractions, whereas the yields of C₇ and C₈ aromatic fractions are increased due to the enhancement of heavy olefin cracking into light olefins (C₂–C₄) and secondary condensation reactions of these light olefins to yield aromatics. Hence, the yield of the BTX fraction increases significantly by reducing the SiO₂/Al₂O₃ ratio of the zeolite from 280 to 30: the yield of benzene from 0.12 to 1 wt % and those of toluene and xylenes from 0.56 and 0.68 wt % to 4 wt %.

Table 3 compares the fuel properties of the gasoline fractions obtained using the three HZSM-5 zeolite based catalysts. As observed, an increase in the acidity of the catalyst leads to a product with a higher octane number, with the content of aromatics and benzene being higher and olefin concentration lower. Olefin content exceeds in all cases the maximum established by the UE to avoid polymerization problems. Furthermore, the octane number is lower than the one of commercial gasoline (95 or 98). However, the absence of sulfur makes this fraction suitable for blending it with refinery streams that surpass the upper limit of 10 ppm for this component.

The results obtained evidence that, although the catalyst has been subjected to a severe thermal stabilization (calcination at

575 °C) and undergoes a significant dehydroxylation that reduces the acid strength and the Brønsted/Lewis ratio in the three catalysts based on HZSM-5 zeolite,³⁵ the kinetic behavior highly depends on the acid structure. Consequently, the acid properties of the zeolite are a key factor to optimize product distribution toward a commercially interesting product stream with high propylene content, thus contributing to meeting its growing demand.⁵¹ Apart from the modification of the SiO₂/Al₂O₃ ratio, many methods have been studied in the literature for modifying the acid structure of the HZSM-5, such as the incorporation of alkaline or transition metals (P, La, Ca or Ni),^{52–54} or the application of other treatments (desilication, dealumination, passivation, or silylation) that also alter the porous structure and shape selectivity of the catalyst.^{55–57}

3.3. Comparison of Pyrolysis-Catalytic Cracking (Two-Step Unit) with Catalytic Pyrolysis (Single-Step Unit) for Olefin Production. On the basis of the aforementioned results, the characteristics of the reaction system used give way to higher olefin selectivities than in other studies in which the catalytic pyrolysis of HDPE was carried out on HZSM-5 zeolites in a fluidized-bed reactor.^{8,9,11} Thus, Hernandez et al.⁸ studied the effect of residence time in the pyrolysis of HDPE over a HZSM-5 zeolite in a fluidized-bed reactor. They observed that high yields of propylene and isobutene are obtained when operating with short residence times, whereas with an increase in residence time the yield of aromatics, such as benzene, is higher due to the enhancement of secondary reactions. Similarly, Mastral et al.⁹ studied the pyrolysis of HDPE in a fluidized-bed reactor using a HZSM-5 catalyst, and they obtained the maximum light olefin yield at 450 °C, 43.6 wt %, with butenes being the most abundant olefins. Lin and Yang¹¹ studied the batch pyrolysis of a mixture of plastic wastes on a HZSM-5 zeolite in a fluidized-bed reactor. They observed a maximum C₁–C₄ fraction yield of 61 wt % at 430 °C.

The studies involving two-steps (i.e., pyrolysis and subsequent catalytic cracking) are scarce and have involved batch mode.^{31,58} San Miguel et al.³¹ studied the batch pyrolysis of waste polyethylene and the inline cracking of the volatiles formed on a HZSM-5 zeolite. These authors have observed an increase in C₁–C₄ fraction as cracking temperature is raised, with the maximum yield being 53 wt % at 475 °C. Bagri and Williams⁵⁸ studied a two-step process consisting of two fixed-bed reactors. The effect of cracking temperature on a HZSM-5 zeolite was studied in a wide temperature range of 400 to 600 °C, with the highest C₁–C₄ fraction yield of around 27 wt % being obtained at 600 °C.

Elordi et al.^{14–16} studied in situ catalytic pyrolysis of HDPE in a conical spouted bed reactor using the same HZSM-5 zeolite-based catalysts with the aim of maximizing light olefin yield. However, this single-step strategy has several limitations involving process conditions when operation occurs in continuous mode. Thus, the bed mass (catalyst or catalyst + sand) used in the CSBR is limited by the hydrodynamic conditions required for suitable heat and mass transfer rates and to avoid defluidization problems. Accordingly, the catalyst space time (15–30 g_{cat} min g_{HDPE}^{–1}) was 2–4 times higher than the one used in this study (8 g_{cat} min g_{HDPE}^{–1}) because the fixed bed catalytic reactor (second step) is not limited by the hydrodynamics of the CSBR.

The in situ catalytic pyrolysis process takes place through the following steps: (i) plastic melting; (ii) catalyst particles coating with fused plastic; (iii) fused plastic pyrolysis; and (iv) catalytic transformation of fused plastic and pyrolysis

volatiles.¹⁴ In the case of the two-step unit proposed in this study, the physical and thermal processes (melting, coating, and thermal pyrolysis) are carried out in a conical spouted bed reactor with sand as bed material, whereas the catalytic process takes in a different reactor, which means that pyrolysis and catalytic cracking reactions do not compete on the same acid sites. Therefore, the two-step process has clear operational and scaling-up advantages. First, the temperature in each step can be independently optimized, whereas in the single-step process the minimum advisable temperature is 500 °C in order to avoid defluidization problems involving catalyst particles agglomeration with fused plastic. Second, given that both pyrolysis and catalytic cracking are endothermic reactions, the two-step process facilitates heat transfer. Finally, the cracking capacity of the catalyst in the fixed-bed reactor is not conditioned by the physical constraints of the pyrolysis step and the feed into the fixed-bed reactor is homogeneous (formed mainly by waxes). Consequently, it is possible to operate the fixed-bed reactor with lower space times (lower catalyst masses) than in the single-step process.

Furthermore, the fixed bed catalytic reactor is easier to control, and suitable ranges may therefore be established for operating variables (space time and temperature), thus attenuating secondary reactions thanks to higher catalyst efficiency and controlled operating conditions. The development of a kinetic model will allow for both processes simulation and studies aimed at scaling-up. Ideal plug flow is a reasonable assumption in fixed-bed reactors, whereas single-step modeling is a challenge because polyolefin cracking kinetics depends on the physical steps (heating, melting, and subsequent pyrolysis of the plastic), as well as on the gas and solid flow pattern in the spouted bed reactor.

Regarding the application of this process to the polyolefins contained in the municipal waste, the first pyrolysis step in the two-step process allows separation of the inorganic additives in the commercial plastics, given they cause operating problems in the catalytic reactor. Furthermore, the arrangement of a cracking reactor in line with a first one for pyrolysis allows the resolution of crucial problems, such as catalyst deactivation; that is, parallel reactors operating in reaction–regeneration cycles may be used.

Table 4 compares the results obtained by Elordi et al.¹⁶ by in situ catalytic pyrolysis using catalysts based on HZSM-5 zeolite with SiO₂/Al₂O₃ ratios of 30 and 80 with those obtained here in the two-step reaction system (Figure 8) at the same operating temperature (500 °C). In both cases, the main product fraction corresponds to light olefins (C₂–C₄), with propylene being the individual compound with the highest yield.

As observed, the yields obtained for light olefins and propylene are very similar in both reaction systems, although those for the single-step have been obtained with a higher space time. However, the effect of catalyst acidity (SiO₂/Al₂O₃ ratio) is different, as for the in situ catalytic pyrolysis the yields of light olefins and propylene peak for the catalyst with a SiO₂/Al₂O₃ ratio of 80, whereas in the two-step process of pyrolysis and catalytic cracking the most acidic catalyst (SiO₂/Al₂O₃ ratio of 30) gives way to the highest production of light olefins and propylene.

In the two-step process of pyrolysis and catalytic cracking, in which the yields of products formed by secondary reactions are low, an increase in catalyst acidity gives way to a more severe cracking of the heavy fractions and, therefore, a higher yield of

light olefins. Nevertheless, the yield of light olefins in the in situ catalytic pyrolysis decreases slightly when the process includes a more acidic catalyst, as olefin condensation reactions to produce aromatics are enhanced because of the higher catalyst amount used.

CONCLUSIONS

The process of pyrolysis and downstream catalytic cracking is an interesting technology for light olefin production from high-density polyethylene. The separation of the physical and catalytic steps provides a higher versatility as it allows optimization of the operating conditions in each step and, therefore, an increase in the efficiency compared to the alternative technology of in situ catalytic pyrolysis.

HDPE pyrolysis is carried out in a conical spouted bed reactor at 500 °C with no defluidization problems, thanks to the excellent features of this reactor for minimizing the limitations of the melting–coating–devolatilization stages. As a consequence of these characteristics and the low residence time of the volatile stream in the CSBR, the pyrolysis products are composed mainly of waxes (C_{21+}), 67 wt %, which are fully transformed at 500 °C on a HZSM-5 zeolite in a downstream fixed bed catalytic reactor to yield mainly light olefins, thanks to the severe shape selectivity, moderate acid strength, and low hydrogen transfer capacity of this zeolite.

The acidity of the HZSM-5 zeolite has been proven to be an essential parameter to optimize product distribution toward a light olefin stream with high propylene content. Thus, an increase in the catalyst acidity (lower SiO_2/Al_2O_3 ratio) leads to a higher yield of light olefins, due to the higher activity of the catalyst for cracking C_{12} – C_{20} and C_5 – C_{11} fractions. Furthermore, the yields of light alkanes and aromatics are also increased by enhancing the secondary hydrogen transfer and condensation reactions. A high yield of light olefins (58 wt %) is obtained on a HZSM-5 zeolite with a SiO_2/Al_2O_3 ratio of 30, with propylene being the main compound with a yield of 32 wt %.

The low residence time in the catalytic reactor enhances the selectivity of light olefins and attenuates the secondary reactions of coke formation. In addition, the porous structure of the catalyst particles also contributes to the low coke deposition, as the mesopores formed by agglomerating the zeolite with bentonite and alumina favor the diffusion of coke precursors toward the outside of the particle, thus avoiding the blockage of the external micropores of the zeolite.

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Notes

The authors declare no competing financial interest.

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